CIRCLE PACKINGS OF PRESCRIBED PLANAR OPEN SETS

Abstract. Every nonempty open subset of the plane admits a countably infinite circle packing whose tangent-halfplane polygons form a locally finite, face-to-face tiling of precisely that set. Every polygon has at most six sides and lies between its inscribed disk and the concentric disk of twice the radius. The construction consists of two explicit local modifications of the hexagonal tiling, iterated near the complement of the prescribed set.

1. Statement and definitions

Write $B(v,r)$ and $\overline{B}(v,r)$ for the open and closed Euclidean disks. A polygon will mean a bounded, closed, convex polygon with nonempty interior; its edges are its maximal nondegenerate sides. A *locally finite, face-to-face tiling* of an open set $D \subseteq \mathbb{R}^2$ is a family $\{Q_v\}$ such that

$$Q_v \subseteq D, \qquad \bigcup_v Q_v = D, \qquad \operatorname{int} Q_v \cap \operatorname{int} Q_w = \varnothing \quad (v \neq w),$$

each compact subset of D meets finitely many tiles, and each nonempty intersection of distinct tiles is a full edge of both or a vertex of both. Given points $v \in \operatorname{int} Q_v$, join v and w when $Q_v \cap Q_w$ is an edge. This tiling graph is *orthogonal* if the vector $w - v$ is perpendicular to the line containing that edge, as in [1, Section 1.1].

A *circle packing* is a collection $\{\overline{B}(v, r_v) : v \in V\}$ of positive-radius closed disks with pairwise disjoint interiors. Write $v \sim w$ when $|v - w| = r_v + r_w$. For such a pair, the common tangent line bounds the halfplane

$$H^-_{v,w} = \left\{ x \in \mathbb{R}^2 : (x - v) \cdot \frac{w - v}{|w - v|} \leq r_v \right\} \tag{1.1}$$

containing the disk at v. Define its *tangency polygon*, when the intersection is a bounded polygon, by

$$P_v = \bigcap_{w \sim v} H^-_{v,w}. \tag{1.2}$$

The intersection is over *all* tangent disks and is an intersection of halfplanes, not of lines.

Theorem 1.1. *For every nonempty open set $\Omega \subseteq \mathbb{R}^2$ and every $\varepsilon > 0$, there is a countably infinite circle packing, with all its disks contained in Ω, such that the polygons* (1.2) *form a locally finite, face-to-face tiling of Ω. For distinct centers v, w, the disks are tangent if and only if $P_v \cap P_w$ is an edge, and that edge lies in their common tangent line. Consequently, the tangency graph is the orthogonal tiling graph. Each polygon has at most six sides, and*

$$0 < r_v \leq \varepsilon, \qquad \overline{B}(v, r_v) \subseteq P_v \subseteq \overline{B}(v, 2r_v). \tag{1.3}$$

Corollary 1.2. *If $U \subseteq \mathbb{R}^2$ is open, $C \subseteq \mathbb{R}^2$ is closed, and $U \setminus C \neq \varnothing$, then Theorem 1.1 applies to $\Omega = U \setminus C$. No measure, dimension, or countability hypothesis on C is required.*

Coverage here is by the polygons, not by the disks, and the result is only in dimension two. Local finiteness is relative to Ω; accumulation at its boundary is allowed.

2. Tangency polygons and a local refinement

A polygon is *circumscribed* about a disk if it contains the disk and every side line is tangent to it. Two such disks have *matching contacts* across a shared edge if both touch its line at the same point.

Lemma 2.1. *Suppose that a locally finite, face-to-face tiling $\{Q_v\}$ of an open set $D \subseteq \mathbb{R}^2$ has circumscribed disks $\overline{B}(v, r_v)$, with matching contacts across every shared edge. Then these disks form a circle packing; their tangencies are exactly the edge adjacencies; and $Q_v = P_v$ for* (1.2).

Proof. Each open disk lies in its polygon's interior, so the disk interiors are disjoint. A disk contained in a polygon cannot meet a polygon vertex: the two incident side lines would be distinct supporting lines of the disk at the same point, whereas a circle has a unique supporting line there.

If two disks are tangent at t, then t lies on both polygon boundaries. Indeed, if it were interior to one polygon, interior points of the other polygon arbitrarily near t would give overlapping interiors. Thus t lies in the relative interior of an edge of each polygon, and the face-to-face property makes these the same edge. Conversely, matching contacts across an edge give disk tangency.

Every edge E of a tile has a neighboring tile. To see this, take t in its relative interior and points outside the tile converging to t in a compact neighborhood contained in D. Coverage and local finiteness give a subsequence in a fixed other tile. Its closedness and the face-to-face property show that its intersection with the original tile is E. Hence the halfplanes in (1.2) are precisely the side halfplanes of Q_v, whose intersection is Q_v. The center-to-center line is perpendicular to each common tangent line. □

Set

$$a = (2, 0), \quad b = (1, \sqrt{3}), \quad \Lambda = \mathbb{Z}a + \mathbb{Z}b, \quad n_k = (\cos(k\pi/3), \sin(k\pi/3)), \quad 0 \le k < 6, \tag{2.1}$$

with subscripts modulo six, and let

$$H = \{x : n_k \cdot x \le 1 \text{ for all } k\}, \qquad F_k = H \cap \{x : n_k \cdot x = 1\}. \tag{2.2}$$

The vertices of H are $(1, \pm 1/\sqrt{3})$, $(-1, \pm 1/\sqrt{3})$, and $(0, \pm 2/\sqrt{3})$. Its inradius is 1 and its circumradius is $2/\sqrt{3}$.

The polygons $z + H$, $z \in \Lambda$, form the regular hexagonal tiling with matching disks $\overline{B}(z, 1)$. For completeness, a lattice vector $ma + nb$ has squared length $4(m^2 + mn + n^2)$. The six shortest nonzero vectors are $2n_k$; all others have length at least $2\sqrt{3}$, since $m^2 + mn + n^2$ is never 2 modulo 3. For $x \in H$ and any other nonzero lattice vector λ,

$$x \cdot \lambda \le (2/\sqrt{3})|\lambda| \le |\lambda|^2/2.$$

Thus H is the lattice Voronoi cell: its six inequalities are exactly the nonredundant nearest-point inequalities. The resulting lattice cells cover the plane and meet face-to-face. Across $z + F_k$, the disks centered at z and $z + 2n_k$ touch at $z + n_k$.

For $R > 0$, write $\Lambda_R = R\Lambda$. The *standard radius-R pair at $z \in \Lambda_R$* is

$$\big(\overline{B}(z, R),\ z + RH\big).$$

Lemma 2.2 (Local refinement)**.** *Let $S \subseteq \Lambda_R$. Suppose a whole-plane tiling satisfies the hypotheses of Lemma 2.1 and contains the standard radius-R pair at each $z \in S$. These pairs can be replaced, leaving every other old pair unchanged, to give another such tiling. Every new disk has radius $r = R/3$, and every new polygon Q with center c has at most six sides and satisfies*

$$\overline{B}(c, r) \subseteq Q \subseteq \overline{B}(c, 2r).$$

All new polygons are contained in $\bigcup_{z \in S}(z + RH)$.

Moreover, for $c \in \Lambda_{R/3}$, if every $z \in \Lambda_R$ with $|z - c| < 3R$ belongs to S, then the resulting tiling contains the standard radius-$R/3$ pair at c.

Proof. We give the construction at $R = 1$; translation and scaling give the general case.

First operation: subdivide a selected hexagon. Replace H, together with its unit disk, by the central pair with polygon $H/3$ and center 0, and six pairs with

$$T_k = \operatorname{conv}\big(F_k \cup (F_k/3)\big), \qquad c_k = \tfrac{2}{3}n_k. \tag{2.3}$$

All seven disks have radius $1/3$. The polygons partition H, as is seen by intersecting rays from the origin with $H/3$ and then the corresponding trapezoid. Their intersections are full edges or vertices. For example,

$$T_0 = \{(x,y) : 1/3 \le x \le 1,\ |y| \le x/\sqrt{3}\}. \tag{2.4}$$

The distances from $(2/3, 0)$ to its four side lines are $1/3$. The central disk is inscribed in $H/3$. The center distances between the central disk and a peripheral disk, and between consecutive peripheral disks, are $2/3$. If q is the common endpoint of F_k, F_{k+1}, the latter contact is $q/2$, in the relative interior of the edge $[q/3, q]$.

The peripheral disk at c_k meets the old side at n_k. An outside disk of radius ρ formerly meeting that side has center $(1+\rho)n_k$, by matching contacts, so the new disk is tangent to it at the same point. If both old cells are selected, the replacement centers across this side are $(2/3)n_k$ and $(4/3)n_k$, again with contact n_k. No old boundary edge is subdivided.

The vertices $(1, \pm 1/\sqrt{3})$ and $(1/3, \pm 1/(3\sqrt{3}))$ of T_0 have distances $2/3$ and $2/(3\sqrt{3})$, respectively, from $(2/3, 0)$. Convexity gives the asserted shape bound; the central hexagon satisfies it as well.

Second operation: insert a disk at a selected old vertex. Call an old lattice-tiling vertex q eligible when all three incident old hexagons were selected. Its six incident peripheral centers are

$$q + \tfrac{2}{3}n_k, \qquad 0 \le k < 6.$$

Insert the radius-$1/3$ disk centered at q, with polygon $q + H/3$, cutting off one corner of each of those six trapezoids.

Here is the exact polygonal replacement. At the upper outer vertex $q_+ = (1, 1/\sqrt{3})$ of T_0, keep

$$T_0 \cap \{x + \sqrt{3}y \le 4/3\}. \tag{2.5}$$

The closure of the removed corner is the triangle with vertices

$$(1, 1/\sqrt{3}), \qquad (1, 1/(3\sqrt{3})), \qquad (2/3, 2/(3\sqrt{3})). \tag{2.6}$$

The last two are consecutive vertices of $q_+ + H/3$. Rotations by $2\pi/3$ about q_+ and reflection in $x = 1$ give the six center-to-side triangles of that hexagon, so the removed triangles partition it. Each cutting line is the common tangent line of the retained peripheral disk and the inserted disk. Thus the polygons still cover exactly the same region. The portions removed from the two sides of each shared old edge agree, and every new edge is a full side of the inserted hexagon. This proves the face-to-face property.

Each old peripheral disk lies in the retained halfplane of every cut. Hence its old contact points remain, and none can become a polygon vertex by the supporting-line argument in Lemma 2.1. Old shared edges therefore retain positive length and matching contacts. New shared edges have matching contacts with the inserted disk.

The old vertices form the two cosets $\pm(a+b)/3 + \Lambda$. Within one coset their separation is at least 2. Between the two cosets, squared distances have the form

$$\frac{4}{9}(u^2 + uv + v^2), \qquad u, v \in \mathbb{Z}, \quad u \equiv v \equiv 2 \pmod 3.$$

The parenthesis is a positive multiple of 3, so distinct old vertices are at distance at least $2/\sqrt{3}$. This is greater than twice the circumradius $2/(3\sqrt{3})$ of an inserted polygon. The inserted polygons are therefore disjoint and the cuts commute.

Each inserted polygon lies in its three selected old cells. An edge between a selected and an unselected old cell has no eligible endpoint, so it is not subdivided by these cuts. No unselected old pair is altered, even on its boundary. Each trapezoid is cut at at most its two outer corners, so the new polygons have at most six sides. Truncation preserves the inscribed disk and the outer-radius bound.

If both outer vertices of T_0 are eligible, the result is exactly

$$T_0 \cap \{x + \sqrt{3}y \le 4/3\} \cap \{x - \sqrt{3}y \le 4/3\} = (2/3, 0) + H/3. \tag{2.7}$$

Indeed, the four inequalities in (2.4) and these two additional inequalities are precisely the six side inequalities of the hexagon on the right.

Which smaller standard pairs are produced. The nine classes of $\Lambda/3$ modulo Λ have representatives

$$0, \quad \pm a/3, \quad \pm b/3, \quad \pm(a-b)/3, \quad \pm(a+b)/3. \tag{2.8}$$

The middle six give the uniquely represented peripheral points $z + (2/3)n_k$. The last two classes are exactly the old tiling vertices, by the listed coordinates of H.

For an old center, selecting its own cell produces the central standard pair. For an old vertex, selecting its three incident cells produces the inserted standard pair; their centers are at distance $2/\sqrt{3} < 3$. For a peripheral point $c = z + (2/3)n_k$, it is enough to select

$$z, \qquad z + 2n_k, \qquad z + 2n_{k-1}, \qquad z + 2n_{k+1}. \tag{2.9}$$

These are the centers incident to either endpoint of $z + F_k$, and all lie within distance $2 + 2/3 < 3$ of c. Both endpoints are eligible, so (2.7) produces the full smaller hexagon. Further selections cannot alter these full pairs: central hexagons are never cut, inserted hexagons are disjoint, and a peripheral trapezoid has only the two possible corner cuts just described.

Finally, the operations preserve local finiteness. Each new polygon is contained in the union of the selected old cells incident to it, and only finitely many new polygons are associated with any one old cell. A compact set meets finitely many old cells and therefore finitely many new polygons. This completes the proof. □

3. Refinement near the complement

Fix Ω and ε as in Theorem 1.1, and let $F = \mathbb{R}^2 \setminus \Omega$. If $F = \varnothing$, the standard radius-ε packing proves the theorem. Henceforth $F \neq \varnothing$. Set

$$R_n = \varepsilon 3^{-n}, \qquad \Lambda_n = R_n\Lambda, \qquad \mathcal{N}_t(F) = \{x : \operatorname{dist}(x, F) < t\}.$$

We construct whole-plane tilings $\mathcal{T}_n$ by disk–polygon pairs satisfying Lemma 2.1, with all radii in $[R_n, \varepsilon]$, at most six sides per tile, and the shape bound (1.3). We also maintain the invariant

$$z \in \Lambda_n, \quad \operatorname{dist}(z, F) \le 12R_n \quad \Longrightarrow \quad \big(\overline{B}(z, R_n), z + R_nH\big) \in \mathcal{T}_n. \tag{3.1}$$

This is agreement of *complete* disk–polygon pairs, not agreement only on a neighborhood of F.

Start with the standard radius-R_0 tiling. Given $\mathcal{T}_n$, apply Lemma 2.2 with $R = R_n$ and

$$S_n = \{z \in \Lambda_n : \operatorname{dist}(z, F) \le 8R_n\}. \tag{3.2}$$

All required old pairs are available by (3.1). The result is $\mathcal{T}_{n+1}$. For $c \in \Lambda_{n+1}$ with $\operatorname{dist}(c, F) \le 12R_{n+1}$, every $z \in \Lambda_n$ with $|z - c| < 3R_n$ satisfies

$$\operatorname{dist}(z, F) < (4 + 3)R_n = 7R_n < 8R_n.$$

Thus the last assertion of Lemma 2.2 proves (3.1) at stage $n + 1$. The other inductive properties follow from its first assertion.

Crucially, the construction has the following support property:

$$\begin{gathered}\text{Every old pair removed, and every new pair introduced, in } \mathcal{T}_n \to \mathcal{T}_{n+1} \\ \text{has its entire polygon contained in } \mathcal{N}_{10R_n}(F).\end{gathered} \tag{3.3}$$

Indeed, only selected old pairs change; their polygons have centers at distance at most $8R_n$ from F and circumradius $2R_n/\sqrt{3} < 2R_n$. Every new polygon lies in their union. Unselected old pairs remain exactly unchanged.

Every finite stage is locally finite on the whole plane by Lemma 2.2, and is therefore countable by exhaustion with closed Euclidean balls. Neither F nor S_n is required to be bounded or finite.

4. The limiting tiling

Retain exactly the eventually unchanged disk–polygon pairs:

$$\mathcal{T}_\infty = \bigcup_{N\geq 0} \bigcap_{n\geq N} \mathcal{T}_n. \tag{4.1}$$

Removed disks are not retained merely because their centers occurred at an earlier stage.

Lemma 4.1. *For each compact $K \subseteq \Omega$, the pairs whose polygons meet K are identical in $\mathcal{T}_\infty$ and in $\mathcal{T}_N$ for all sufficiently large N. They are finite in number and cover K.*

Proof. For nonempty K, compactness and closedness give $\delta = \operatorname{dist}(K, F) > 0$. Choose N such that $10R_N < \delta$. For every $n \geq N$, (3.3) prevents both the removal of a pair meeting K and the introduction of a new pair meeting K. Hence the complete collection of pairs meeting K remains unchanged. Finite-stage local finiteness and coverage give the conclusion. The empty case is immediate. □

Lemma 4.2. *No polygon of $\mathcal{T}_\infty$ meets F.*

Proof. We prove that every pair of $\mathcal{T}_n$ whose polygon meets F is removed in the next step. Suppose $x \in Q \cap F$ for a tile Q of $\mathcal{T}_n$. Every standard radius-R_n hexagon meeting $B(x, R_n)$ has its center within distance

$$R_n + 2R_n/\sqrt{3} < 3R_n$$

of x. By (3.1), all these full pairs occur in $\mathcal{T}_n$, and their polygons cover $B(x, R_n)$. Since Q is full-dimensional and convex, $\operatorname{int} Q \cap B(x, R_n)$ is nonempty and open, including when x is a vertex of Q. Finitely many standard hexagon edges cannot cover this open set. Hence the interior of Q meets the interior of one of these standard cells, and disjointness of tile interiors forces the two disk–polygon pairs to coincide. Thus $Q = z + R_n H$ for a standard pair. Because $x \in Q \cap F$, its center satisfies

$$\operatorname{dist}(z, F) \leq |z - x| \leq 2R_n/\sqrt{3} < 8R_n.$$

This pair is selected and replaced by pairs of radius $R_n/3$, so it is absent from $\mathcal{T}_{n+1}$. An eventually unchanged pair therefore cannot meet F. □

Proof of Theorem 1.1. Any two distinct retained pairs coexist in a sufficiently late finite stage. Hence they have distinct centers, their polygons have disjoint interiors and meet face-to-face, and their disks have matching contacts whenever the polygons share an edge. Index the retained pairs by their centers as $(\overline{B}(v, r_v), Q_v)$. Each inherits its circumscribed disk, positive radius, shape bound, and side bound from a finite stage.

Lemmas 4.1 and 4.2 show that these polygons form a locally finite tiling of exactly Ω. Lemma 2.1 therefore identifies Q_v with (1.2) and proves all the tangency and orthogonality assertions.

Each stage is countable, so (4.1) is countable. It is infinite: a finite union of bounded closed polygons is compact, whereas the nonempty open set Ω is not. This proves the theorem. □

Proof of Corollary 1.2. The set $U \setminus C$ is open, with closed complement $(\mathbb{R}^2 \setminus U) \cup C$. Apply Theorem 1.1. □

References

[1] Ahmed Bou-Rabee and Ewain Gwynne, *Random walk on sphere packings and Delaunay triangulations in arbitrary dimension*, arXiv:2405.11673v2, 2025.